

Privacy Risk & Utility Assessment Report

Healthcare Claims Analytics Platform

Phase 7 – Privacy & De-Identification Evaluation

1. Executive Summary

This report presents the privacy risk assessment and utility validation results for the synthetic healthcare claims analytics platform designed to identify avoidable Emergency Room (ER) utilization and quantify cost optimization opportunities.

The objective of Phase 6 was to evaluate re-identification and attribute disclosure risks while ensuring that predictive performance and operational metrics remain intact.

Privacy evaluation included:

- k-anonymity (identity disclosure risk)
- l-diversity (attribute disclosure risk)
- t-closeness (distributional disclosure risk)
- Member-level equivalence class validation
- Utility preservation assessment (Model AUC, ER rate stability, risk segmentation stability)

Key Findings

- Minimum claim-level $k = 570$
- Minimum member-level $k = 186$
- No equivalence classes below $k = 10$
- l-diversity satisfied for all evaluated sensitive attributes
- No homogeneity risk detected
- Maximum dominance ratio = 0.707
- t-closeness maximum deviation = 0.371
- No degradation in predictive features or business KPIs

Conclusion

The dataset satisfies de-identification standards appropriate for analytics use. Privacy safeguards were validated without materially degrading predictive performance or operational metrics.

2. Re-Identification Risk Assessment

2.1 Quasi-Identifiers Evaluated

The following attributes were treated as quasi-identifiers:

- Age (binned categories)
- Gender

- Region
- Diagnosis group

These attributes were assessed for linkage vulnerability and equivalence class size.

2.2 k-Anonymity (Identity Disclosure Protection)

- Claim-level minimum k: 570
- Member-level minimum k: 186
- No equivalence class < 10 members

All equivalence classes exceed commonly accepted de-identification thresholds ($k \geq 10$). This significantly mitigates re-identification risk from potential linkage attacks.

2.3 Attribute Disclosure Protection

Sensitive attributes evaluated:

- risk_level
- behavioral_flag
- admission_flag

Results:

- I-diversity (risk_level) = 3
- I-diversity (behavioral_flag) = 2
- I-diversity (admission_flag) = 2

All equivalence classes contain multiple sensitive attribute values, eliminating homogeneity risk.

2.4 Distributional Disclosure (t-Closeness)

- Maximum dominance ratio = 0.707
- Maximum t-closeness deviation = 0.371

While moderate distributional skew exists due to natural epidemiologic variation (e.g., higher risk concentration in older age groups), no deterministic inference risk is present.

3. Mitigation Techniques Implemented

Privacy protection was achieved through structured dataset design rather than post-hoc masking.

Controls applied:

1. Age binning into coarse ranges
2. Aggregated diagnosis grouping (no granular codes)
3. Regional grouping (macro-regions)
4. Removal of direct identifiers
5. No raw service dates released

6. Member-level equivalence validation

No suppression or record removal was required because all equivalence classes exceeded $k \geq 10$ thresholds.

4. Utility Impact Assessment

Privacy validation included formal assessment of analytic utility preservation.

4.1 Model Performance (AUC)

The predictive model was retrained and evaluated using identical feature definitions and split parameters.

Observed AUC range: ~0.63–0.69 across synthetic regenerations.

The difference in AUC values was attributable to synthetic dataset regeneration following additional stochastic cost simulation (`np.random.uniform`). This altered the pseudorandom number stream and produced a statistically different synthetic population.

No modelling features were altered during privacy evaluation.

Conclusion: No predictive degradation attributable to privacy safeguards.

4.2 ER Rate Stability

- ER utilization rate unchanged
- Avoidable ER labelling unchanged
- No distortion in operational KPIs

Conclusion: Business metrics preserved.

4.3 Risk Segmentation Stability

- Distribution of Low / Medium / High risk unchanged
- No category suppression or re-binning introduced
- No segmentation drift observed

Conclusion: Risk stratification integrity maintained.

5. Recommendations

1. Preserve frozen train/test indices for reproducibility.
2. Isolate random number streams in synthetic data generation to prevent random stream drift.
3. Maintain coarse quasi-identifier structure for analytic releases.
4. Continue monitoring dominance ratios if additional sensitive attributes are introduced.
5. If expanding to real-world data, conduct formal HIPAA Safe Harbor and Expert Determination review.

6. Conclusion

The synthetic healthcare claims dataset satisfies identity and attribute disclosure safeguards through strong k-anonymity, l-diversity, and controlled distributional deviation.

Privacy validation did not degrade predictive performance or operational analytics.

The dataset is suitable for de-identified analytic use within the defined scope of this platform.